

MAO-JAN LIN

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RESEARCH INTERESTS AND TECHNICAL SKILLS

Research interests: Pangenomics, immunogenomics, sequence alignment, variant calling, genome assembly

Programming languages: Python, C++, R **Tools:** Git, Bash, Snakemake, LaTeX, Cursor, Codex

EDUCATION

Johns Hopkins University (JHU), Ph.D. candidate, Computer Science 08/2021–12/2026 (Expected)
Advisor: Dr. Ben Langmead

National Taiwan University (NTU), M.S. in Electronics Engineering 09/2016–06/2019
Advisor: Dr. Yi-Chang Lu

- Thesis: “A Parallel Design of Dynamic Programming Sequence Aligner with Affine Gap Traceback”

NTU, B.S. in Electrical Engineering 09/2012–06/2016

RESEARCH EXPERIENCE

Langmead Lab, JHU 08/2021–present
Ph.D. Researcher

- Developed a pangenome-based assembly patching method that produced 69–87% fewer errors and 9–23 Mb greater benchmark coverage than a single-reference patching tool across five assembly datasets [1].
- Implemented a variant-calling module for imputed personalized references, supporting workflows with 6.1–15.0% fewer calling errors compared to pangenome approaches in GIAB high-confidence regions [2].
- Analyzed immunoglobulin (IG) somatic recombination and reduced associated assembly artifacts, recovering an average of 10 additional IGH genes per individual versus HPRC assemblies across 47 samples [3].
- Developed methods to measure and categorize reference bias in sequence alignment [4].
- Mentored an undergraduate in designing experiments to assess how pangenome size affects alignment accuracy.

NTU Hospital, Lab of Dr. Pei-Lung Chen 07/2020–07/2021
Research Assistant

- Built immune-receptor genotyping and annotation pipelines, achieving 100% TRV/TRJ allele-call concordance with assembly-derived annotations in two GIAB samples [5].

NTU, Lab for Data Processing Systems 09/2016–06/2019
M.S. Researcher

- Designed ASIC and CPU–ASIC hybrid accelerators for sequence alignment and variant calling.

INDUSTRY EXPERIENCE

Roche, Computational Biology Intern, Molecular Lab Applications 06/2024–08/2024

- Designed and benchmarked a structural variant calling pipeline for the Sequencing by Expansion (SBX) platform.

Apple, Product Design Intern, Input Device Team 01/2018–07/2018

- Developed manufacturing analysis software and Python serial interfaces for embedded keyboard prototypes.

Himax, Summer Intern, Video Image Processing Team 07/2017–08/2017

SELECTED PUBLICATIONS AND PREPRINTS

1. Mao-Jan Lin, V. S. Shivakumar, B. Langmead, Human Pangenome Reference Consortium, “**ImpuT2T: Pangenome-Based Patching for Human Genome Assemblies**,” *bioRxiv*, 2026 (preprint).
2. K. Vaddadi, Mao-Jan Lin, S. Majidian, T. Mun, B. Langmead, “**Minimizing reference bias with an imputed personalized reference**,” *Genome Research*, 2026.
3. Mao-Jan Lin, B. Langmead, Y. Safonova, “**IGLoo enables comprehensive analysis and assembly of immunoglobulin heavy-chain loci in lymphoblastoid cell lines using PacBio high-fidelity reads**,” *Cell Reports Methods*, 2025.
4. Mao-Jan Lin, S. Iyer, N.-C. Chen, B. Langmead, “**Measuring, visualizing and diagnosing reference bias with biastools**,” *Genome Biology*, 2024.
5. Mao-Jan Lin*, Y.-C. Lin*, N.-C. Chen*, et al., “**Profiling Genes Encoding the Adaptive Immune Receptor Repertoire with gAIRR Suite**,” *Frontiers in Immunology*, 2022. *Co-first authors.